COMPARATIVE ANALYSIS OF VARIOUS WIRELESS TECHNOLOGIES IN RURAL AREAS

Technology	Range	Speed	Rural Factor
Wi-Fi	0.05-0.45	600mbps	Range
WiLD	100-280	3mbps - 4mbps	Stability, cost
Wi-Max	0.3-49	35mbps - 70mbps	Cost, complexity
Satellite	Not related	5mbps - 25mbps	Costly
TVWS (UHF)	10-30	2mbps - 45mbps	Spectrum sensing

over networks and ICT applications. This, along with community antennae, optical fibres, satellites and fixed mobile wireless technologies, can also be used in rural areas effectively. The ability to access the Internet can bring a positive impact on the rural society. However, there is a wide digital divide between urban and rural areas in India, because of uneven distribution of basic telecom infrastructure.

Further, poverty and lack of education are also factors responsible for the lack of advancements in wireless technology in rural areas. Social interaction can be obtained between urban and rural masses through social media channels such as Facebook, Twitter, WhatsApp and the like. Wireless connectivity in rural areas can also reduce poverty, create jobs, and increase skills and income of the population.

To overcome the lack of Internet connectivity for undeserved communities, Digital Employment Foundation and Internet Society jointly launched Wireless for Communities (W4C) initiative in October 2010. This project includes training rural communities on different wireless technologies, among others.

Under Digital Village initiative announced in February 2017, the government of India aims to bring free Wi-Fi to 1050 villages. The initiative will follow the public-private partnership mode of operation. Services including LED streetlighting, Wi-Fi hotspots, skill development, interactive sessions with experts, government officials and so on will be provided at village *panchayat* level.

In the next three years, Narendra Modi government plans to lay down infrastructure including one million kilometres of optical-fibre cable network, to bring digital connectivity to the country's 223,000 villages.

On November 13, 2017, the government announced the second and final phases of BharatNet project, which aims to provide broadband connectivity across Indian villages. It aims to provide Internet services to rural masses at affordable prices. This ambitious project is likely to add US$ 68 billion to India's GDP.

Despite these initiatives, digital access remains abysmal in Indian villages.

Creating and deploying new applications

Union Budget 2018 had good news for e-commerce startups in rural India. The government of India is planning to set up 500,000 Wi-Fi hotspots that are expected to provide broadband Internet access to 50 million people in rural parts of the country.

Wi-Fi hotspots. Wireless hotspots are wireless access points located typically in public locations that provide Internet access to mobile devices such as laptops and smartphones. Typical Wi-Fi hotspot venues include cafés, libraries, airports and hotels. Wi-Fi hotspots can be added to rural areas as well.

In 2005, doctors from Arvind Eye Care Systems in Tamil Nadu used a long-distance Wi-Fi network to interview and diagnose their rural patients. Till date, they have remotely diagnosed 100,000 patients and restored eyesight to 30,000.

Satellites or wireless technologies such as Wi-Max can be used to create rural links. Next-generation wireless Internet technology for rural areas is underway, with point-to-multipoint communications and integrated charge controllers for solar panels. A combination of special antennae and software can be used to convert short-distance Wi-Fi to long-distance Wi-Fi. The result is high bandwidth, in the range of 18mbps to 20mbps.